

Practice problems #7

1. Let $u = \begin{bmatrix} -5 \\ -3 \\ -6 \end{bmatrix}$ and $A = \begin{bmatrix} 3 & 5 \\ 1 & 1 \\ -2 & -8 \end{bmatrix}$.

(a) Let the columns of A be a_1 and a_2 , respectively. Describe geometrically the span of the columns of A .

(b) Fill in the blank and explain: $\text{Span}\{a_1, a_2\}$ is a subset of _____.

(c) Is u in the set spanned by the columns of A ? Why or why not?

(d) Fill in the blank and explain: The vectors a_1, a_2 and u all lie in the *same* _____.

2. Let $a_1 = \begin{bmatrix} -3 \\ 6 \end{bmatrix}$ and $a_2 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$. True or False. Justify:

There is a vector b in $\mathbf{R}^2$ so that the linear system $x_1a_1 + x_2a_2 = b$ is inconsistent.

3. True or False. Justify: The Span of the columns of the matrix $\begin{bmatrix} 0 & 0 & 2 \\ 0 & -5 & 1 \\ 4 & 6 & -3 \end{bmatrix}$ is $\mathbf{R}^3$.

Answers or Hints on page 2.

#1(a): plane in $\mathbf{R}^3$ containing the vectors a_1 and a_2

#1(b): $\mathbf{R}^3$ (A linear combination of vectors in where?)

#1(c): Yes. (Does the vector equation $x_1a_1 + x_2a_2 = u$ have a solution?)

#1(d): plane in $\mathbf{R}^3$ (since u is a linear combination of the vectors a_1 and a_2 , u should lie where?)

#2: True. (What is the Span of the given vectors? Does it cover all of $\mathbf{R}^2$?)

#3: True. How many pivot positions?